Philosophy of Space and Time

Lecture 8: *Galilean Spacetime*

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Today's Plan

Where We Left Off

On Tuesday we introduced Galilean spacetime by stripping away one layer of structure from Newton's full picture:

- Full Newtonian spacetime: simultaneity + temporal metric + spatial metric + affine structure + *persistence of spatial points*
- Galilean spacetime: everything *except* the persistence of spatial points
- Result: no absolute velocity, but absolute acceleration survives as curvature of worldlines

Today: work through the details, using both Maudlin's treatment and Geroch's complementary perspective.

The Galilean Boost

Two Diagrams, One Situation

Diagram A

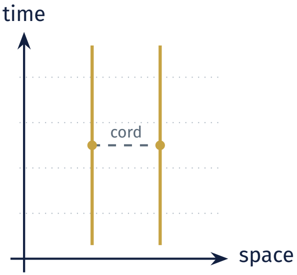
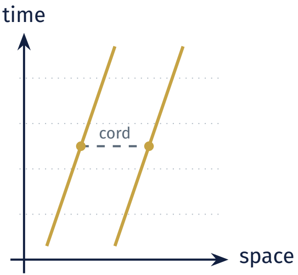


Diagram B (boosted)



In *full* Newtonian spacetime, these depict different situations (rest vs. motion).
In *Galilean* spacetime, they depict the **same physical situation**—because the only difference is absolute velocity.

What a Galilean Boost Preserves

The transformation from Diagram A to Diagram B is a **Galilean boost**: a uniform shearing of spacetime that tilts all worldlines by the same angle.

Preserved by a Galilean Boost

- Elapsed time between events
- Simultaneity (which events are at the same time)
- Spatial distance between simultaneous events
- Which worldlines are straight (inertial)
- Relative velocity of any two bodies

Changed by a Galilean Boost

- Which worldlines are vertical ("at rest")
- The absolute velocity of any body

Symmetries of Galilean Spacetime

The symmetry group of Galilean spacetime includes:

- Spatial translations and rotations
- Time translations
- **Galilean boosts**: adding a uniform velocity to everything

This is a *larger* symmetry group than full Newtonian spacetime. And it *matches* the symmetry group of Newton's dynamics (Corollary V).

?? *Galilean spacetime resolves the mismatch: the spacetime symmetries now agree with the dynamical symmetries. No unobservable structure remains.*

Newton’s Globes Revisited

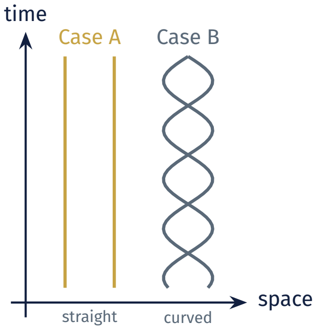
The Globes in Spacetime

Recall Newton's two globes:

- Case A: globes at rest, no tension
- Case B: globes rotating, tension in cord

In the spacetime picture:

- Case A: both worldlines are straight (inertial)
- Case B: the worldlines are helical—they spiral around each other
- A helix is *curved*: this curvature is what produces the tension



How Galilean Spacetime Resolves the Debate

Newton was right:

- Rotation is absolute—it has effects (tension, concavity) that cannot be explained relationally
- Inertia requires non-relational spacetime structure

Leibniz was right:

- Newton postulated more structure than needed
- Absolute position and velocity are surplus
- The shift arguments correctly identified the excess

The Resolution

Galilean spacetime retains the structure Newton *needs* (affine structure for inertia) while discarding what he doesn't (absolute velocity).

Machian Spacetime

If we strip away *even more* structure than Galilean spacetime:

- Keep: simultaneity, Euclidean geometry on each time slice
- Discard: the connection between different time slices entirely (no affine structure)
- Result: **Machian spacetime**—no fact about velocity *or* acceleration
- Sensible questions: how far apart are simultaneous events? Does the distance between two bodies change over time?
- Meaningless: is this body accelerating? Is this body rotating?

Machian spacetime has too *little* structure for Newtonian physics—the bucket experiment has no explanation. But it represents the extreme relationist position.

The Mismatch in the Other Direction

Recall the **mismatch principle**: the symmetries of spacetime should match the symmetries of the dynamics.

- Full Newtonian spacetime: too *little* symmetry (no boosts) for dynamics that *do* have boost symmetry $\rightarrow$ too much structure
- Machian spacetime: too *much* symmetry for dynamics that *lack* that symmetry $\rightarrow$ too little structure

The Machian Mismatch

In Machian spacetime, the allowed transformations include *time-dependent* rotations: $R(t)$ can change from instant to instant. But Newton's dynamics *can* distinguish a rotating frame from a non-rotating one. So the spacetime treats as "the same" situations that the physics treats as different.

Machian Spacetime and the Bucket

In Machian spacetime:

- There is no fact about whether a body is accelerating or rotating
- The bucket experiment distinguishes rotating from non-rotating—but Machian spacetime has no structure to underwrite this distinction
- A time-dependent rotation is a symmetry of the spacetime but *not* a symmetry of the dynamics

This is the mirror image of Newton's problem: where Newton had structure the dynamics couldn't see, Mach would lack structure the dynamics *need*.

Full Newtonian
spacetime structure $>$ dynamics
(too much)

↓

Galilean
spacetime $=$ dynamics
(just right)

↓

Machian
spacetime structure $<$ dynamics
(too little)

Geroch on
Spacetime Structure

Geroch's Approach

Geroch (Chs. 3–4) develops a spacetime approach to this debate:

- Start with **particles** and **observers** as primitive notions
- A particle traces a worldline through spacetime
- An observer is a special kind of particle: one that carries a clock and can make measurements
- The structure of spacetime is revealed by what observers can and cannot determine

Where Maudlin asks "what structure must spacetime have?" Geroch asks "what can observers measure?" (These are complementary approaches that converge.)

Inertial Observers

- An **inertial observer** is one who feels no forces (in “free fall” – not even gravity)
- Distinction between inertial and non-inertial is *physical*, not conventional
- You can *feel* whether you are accelerating (recall the accelerometer from Lecture 6)
- Inertial observers have straight worldlines; non-inertial observers have curved ones

?? *Geroch’s emphasis: the inertial/non-inertial distinction is operationally meaningful. You don’t need to know anything about “absolute space” to determine whether you are accelerating; you just need a sensitive enough instrument.*

Structure of Newtonian Spacetime

In Geroch's framework, Newtonian spacetime provides:

1. A collection of **events**
2. A **time function** assigning elapsed time between events
3. A **spatial distance** between simultaneous events
4. A distinguished class of **straight worldlines** (inertial trajectories)

- Geroch builds the same physics from the spacetime perspective from the outset
- This naturally yields Galilean spacetime rather than full Newtonian spacetime

Reference Frames

Geroch introduces **reference frames** as ways of slicing spacetime into “space at each time”:

- An inertial reference frame $=$ a family of straight, parallel worldlines filling spacetime
- Each worldline represents a “point at rest” in that frame
- In Galilean spacetime, there are infinitely many *equally good* inertial frames, differing by Galilean boosts
- In full Newtonian spacetime, one frame would be singled out as “truly at rest”

?? *A reference frame is not “part of spacetime”—it is a way of describing spacetime. Different frames give different descriptions of the same events, just as different coordinate systems label the same points differently.*

Geroch and Maudlin Compared

Maudlin's Approach

- Start with Newton's ontology (absolute space + time)
- Show the dynamics have more symmetry than the spacetime
- Strip away the excess to get Galilean spacetime
- Top-down: subtract structure

Geroch's Approach

- Start with events and observers
- Ask what structure is operationally accessible
- Build up to Galilean spacetime from below
- Bottom-up: add structure as needed

Both arrive at the same picture. Geroch's approach will prove to be particularly illuminating when we move to relativistic spacetimes.

Break

Inertial Coordinates

What Is an Inertial Coordinate System?

Maudlin (end of Ch. 3) clarifies what “inertial coordinates” actually means:

1. Choose an inertial worldline as the spatial origin
2. On each simultaneity slice, set up Cartesian coordinates on E^3
3. Make the time coordinate proportional to elapsed absolute time

- An inertial coordinate system is *adapted to an inertial frame*
- Newton’s laws take their standard form in such coordinates
- Different inertial coordinate systems are related by **Galilean transformations**

The Galilean Transformations

The Galilean transformations between inertial coordinate systems:

Galilean Transformations

$$\mathbf{x}' = R\mathbf{x} + \mathbf{v}t + \mathbf{a}, \qquad t' = t + b$$

- R: a (constant) rotation of spatial axes
- $\mathbf{v}t$: a **Galilean boost**—the key new ingredient
- $\mathbf{a}$: a spatial translation
- b: a time translation

The crucial point: R is *constant* (not time-dependent). If R could change with time, we would lose the distinction between inertial and accelerated frames—and land in Machian spacetime.

The Classical Debate: A Scorecard

A Hierarchy of Spacetimes

Full Newtonian
absolute rest, velocity, acceleration

too much structure

↓ remove persistence of points

Galilean
no absolute rest/velocity; absolute acceleration

just right

↓ remove affine structure

Machian
no absolute rest, velocity, *or* acceleration

too little structure

The Relativity/Inertia Tension: Resolved?

Recall the central tension from Lecture 5:

Relativity

Uniform motion is undetectable.

Inertia

Straight-line uniform motion is privileged.

Galilean spacetime shows these are *compatible*:

- Inertia requires affine structure (straight vs. curved worldlines)—not absolute velocity
- Relativity eliminates absolute velocity—not affine structure
- The two demands carve out exactly the structure of Galilean spacetime

?? *So the tension between relativity and inertia has a clean resolution in the classical case.*

What Remains of the Newton–Leibniz Debate

- **Newton was right** that physics needs non-relational spacetime structure for inertia and acceleration
- **Leibniz was right** that Newton postulated too much — the kinematic shift arguments correctly identified the surplus
- **Neither fully anticipated** the resolution: Galilean spacetime (formulated only in the 20th century, by Cartan and others)
- **The deeper lesson**: the structure of spacetime is not settled by philosophical arguments alone; it is constrained by what is required to formulate physical theories

Mismatch principle: spacetime symmetries should match dynamical symmetries.

Looking Ahead

What Happens Next?

We have resolved the classical debate using Galilean spacetime. But new questions arise:

- Galilean spacetime still has **absolute simultaneity**: a fact about which events are “at the same time”
- Is this structure also excessive? Could there be physical reasons to give it up?
- Einstein’s answer (1905): yes. The speed of light forces us to abandon absolute simultaneity.
- The result: **Minkowski spacetime**: a different geometry, with combined *spacetime* distance as fundamentsl

The move from Galilean to Minkowski spacetime is in many ways analogous to the move from full Newtonian to Galilean: we strip away structure. But the reasons for doing so are different, and the consequences are far more dramatic.

From Galilean to Minkowski Spacetime: A Preview

Galilean
Absolute simultaneity
Euclidean space
Inertial structure

replace simultaneity
with light cones

Minkowski
No absolute simultaneity
Light-cone structure
Inertial structure

Galilean Spacetime: What Remains to Question?

Galilean spacetime resolves the Newton–Leibniz debate. But it still has substantial structure:

- **Absolute simultaneity**: there is a fact about whether two events happen "at the same time," independent of any observer
- **Separate metrics**: spatial distance (within a time slice) and temporal duration (between slices) are measured by two independent standards
- No upper limit on the speed of causal influence—signals can in principle travel arbitrarily fast

?? *Could there be physical reasons to give up absolute simultaneity—just as Galilean relativity gave us reasons to give up absolute velocity? What kind of evidence would force this?*

The Problem of Light

In the 19th century, Maxwell's theory of electromagnetism introduced a puzzle:

- Maxwell's equations predict that light travels at a definite speed (in a vacuum): $c \approx 3 \times 10^8$ m/s
- Speed relative to *what*? In Galilean spacetime, speed depends on the observer's frame

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- This was identified with the **aether**: a medium filling all of space, at rest in the preferred frame

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The aether would give us absolute velocity back—the very thing Galilean spacetime was designed to eliminate!

An Experimental Question

- If the aether exists and the earth moves through it, the speed of light should differ slightly depending on the direction of measurement (with vs. against the earth's motion)
- Michelson and Morley (1887) designed a precision experiment to detect this difference

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Two Options

The constancy of the speed of light forces a choice:

Option 1: Save Galilean Spacetime

Keep absolute simultaneity and the Galilean boost. Explain away the Michelson-Morley result with additional physical hypotheses (e.g., Lorentz-FitzGerald contraction of the apparatus).

Option 2: Change the Spacetime

Accept that the speed of light really is the same for all inertial observers. Give up absolute simultaneity and the Galilean boost. Replace them with a new spacetime structure.

Einstein (1905) chose Option 2. The result: **Minkowski spacetime**, in which the speed of light is built into the geometry as the light-cone structure.

Summary

1. The **Galilean boost** shears spacetime diagrams without altering the physical content; two boosted diagrams depict the same situation in Galilean spacetime
2. **Newton's globes** survive in Galilean spacetime: rotation $=$ curved worldlines, a geometric fact independent of absolute velocity
3. The **hierarchy** of spacetimes: Full Newtonian $\rightarrow$ Galilean $\rightarrow$ Machian, with Galilean as the "Goldilocks" for classical physics
4. The resolution of the Newton–Leibniz debate: Newton was right about the need for non-relational structure; Leibniz was right about the surplus; Galilean spacetime splits the difference
5. **Geroch's approach**: build spacetime from events and observers; arrive at Galilean structure from below rather than stripping down from above

For Next Time

- **Read:** Maudlin, C 4; Geroch, Chs. 5–6 (Introduction to Spacetime).
- **Think about:**
    - What would it mean to give up absolute simultaneity? What would replace it?
    - How does the speed of light enter the picture?

Up next: **Special Relativity and Minkowski Spacetime**

- The constancy of the speed of light
- Light cones and the causal structure of spacetime
- The Twins Paradox

Overflow: The Galilean Transformations in Detail

In coordinates, the full group of Galilean transformations is:

$$x'^a = R^a{}_b\, x^b + v^a t + c^a, \qquad t' = t + b$$

where $R^a{}_b$ is a *constant* orthogonal matrix, v^a is a constant velocity, c^a is a constant spatial translation, and b is a constant time translation.

Compare with full Newtonian spacetime:

$$x'^a = R^a{}_b\, x^b + c^a, \qquad t' = t + b$$

The only difference: the $v^a t$ term (the boost). Full Newtonian spacetime lacks boost symmetry; Galilean spacetime has it.

Overflow: Goldstein's Implicit Assumption

Maudlin notes that Goldstein's *Classical Mechanics* begins: "Let $\mathbf{r}$ be the radius vector of a particle from some given origin…"

- But what "given origin"?
- If we pick a new origin at each moment, the radius vector changes randomly
- What is really assumed: the origin traces out an *inertial* trajectory
- This presupposes Galilean spacetime (not full Newtonian spacetime)
- Standard treatments of mechanics silently assume Galilean structure—the persistence of absolute space is never used

Overflow: Cartan's Formulation (1923)

- The mathematical formulation of Galilean (neo-Newtonian) spacetime was given by Élie Cartan in 1923
- Cartan showed how to formulate Newtonian gravity as spacetime curvature—anticipating the structure of general relativity but in a non-relativistic setting
- This "Newton-Cartan" theory makes clear that Newtonian gravity can be understood geometrically, without absolute space
- Philosophers of physics (Earman, Friedman) drew on Cartan's work to clarify the Newton–Leibniz debate in the 1970s–80s

Overflow: Does Galilean Spacetime Vindicate Leibniz?

A tempting conclusion: Galilean spacetime shows Leibniz was right all along. But this is too simple:

- Galilean spacetime is *not* a purely relational theory—it still has non-relational structure (affine structure, absolute simultaneity)
- Leibniz wanted all spatial and temporal structure to reduce to relations among bodies
- The globes thought experiment still refutes full relationism, even in Galilean spacetime
- What Galilean spacetime shows is that the debate is *not* a simple binary—there are intermediate positions, and the right one is determined by the physics

Questions?

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